MA1522 Linear Algebra for Computing
L2 Lesson 16



Tan Kai Meng

Semester 1, 2026/27

Eigenvalues and Eigenvectors

Let $\boldsymbol{A}$ be a square matrix of order n, and let $\lambda \in \mathbb{R}^n$.

λ is an **eigenvalue of $\boldsymbol{A}$** if and only if there exists $\boldsymbol{v} \in \mathbb{R}^n$, $\boldsymbol{v} \neq \boldsymbol{0}$, such that

$$\boldsymbol{A}\boldsymbol{v} = \lambda \boldsymbol{v}.$$

If λ is an eigenvalue of $\boldsymbol{A}$, any nonzero vector $\boldsymbol{v} \in \mathbb{R}^n$ such that $\boldsymbol{A}\boldsymbol{v} = \lambda \boldsymbol{v}$ is an **eigenvector of $\boldsymbol{A}$** associated with the eigenvalue λ.

$$\begin{aligned}
\lambda \text{ is an eigenvalue of } \boldsymbol{A} &\Leftrightarrow \mathrm{Null}(\lambda \boldsymbol{I}_n - \boldsymbol{A}) \neq \{\boldsymbol{0}\} \\
&\Leftrightarrow \mathrm{nullity}(\lambda \boldsymbol{I}_n - \boldsymbol{A}) > 0 \\
&\Leftrightarrow \mathrm{rank}(\lambda \boldsymbol{I}_n - \boldsymbol{A}) < n \\
&\Leftrightarrow \lambda \boldsymbol{I}_n - \boldsymbol{A} \text{ is not invertible} \\
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Let $\boldsymbol{A}$ be a square matrix. Determine which of the following statements is/are true:

a. There exists $\lambda \in \mathbb{R}$ such that λ is an eigenvalue of $\boldsymbol{A}$.

b. If λ is an eigenvalue of $\boldsymbol{A}$, then λ is an eigenvalue of $\boldsymbol{A}^T$.

c. If $\boldsymbol{B}$ is row equivalent to $\boldsymbol{A}$ and λ is an eigenvalue of $\boldsymbol{A}$, then λ is an eigenvalue of $\boldsymbol{B}$.



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Solution.

a. Let $\boldsymbol{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Then $\det(\lambda \boldsymbol{I}_2 - \boldsymbol{A}) = \det \begin{pmatrix} \lambda & 1 \\ -1 & \lambda \end{pmatrix} = \lambda^2 + 1 \neq 0$ for all $\lambda \in \mathbb{R}$.

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b. $\det(\lambda \boldsymbol{I}_n - \boldsymbol{A}^T) = \det((\lambda \boldsymbol{I}_n - \boldsymbol{A})^T) = \det(\lambda \boldsymbol{I}_n - \boldsymbol{A})$, so $\det(\lambda \boldsymbol{I}_n - \boldsymbol{A}^T) = 0$ if and only if $\det(\lambda \boldsymbol{I}_n - \boldsymbol{A}) = 0$.

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c. Let $\boldsymbol{A} = \boldsymbol{I}_2$, and let $\boldsymbol{B} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Then $\boldsymbol{B}$ is row equivalent to $\boldsymbol{A}$. 1 is an eigenvalue of $\boldsymbol{A}$ but not an eigenvalue of $\boldsymbol{B}$.

Characteristic Polynomial

Let $\boldsymbol{A}$ be a square matrix of order n.

The **characteristic polynomial of $\boldsymbol{A}$** is $\det(x\boldsymbol{I}_n - \boldsymbol{A})$.

The eigenvalues of $\boldsymbol{A}$ are precisely the roots of the characteristic polynomial, i.e. those $x \in \mathbb{R}$ such that

$$\det(x\boldsymbol{I}_n - \boldsymbol{A}) = 0.$$

Let $\lambda \in \mathbb{R}$ be an eigenvalue of $\boldsymbol{A}$.

The **algebraic multiplicity** of λ is the number of times λ appears as a root of $\det(x\boldsymbol{I}_n - \boldsymbol{A})$.

The **geometric multiplicity** of λ is $\text{nullity}(\lambda\boldsymbol{I}_n - \boldsymbol{A})$.

The **eigenspace associated to** λ, denoted E_λ, is

$$E_\lambda = \{\boldsymbol{v} \in \mathbb{R}^n \mid \boldsymbol{A}\boldsymbol{v} = \lambda\boldsymbol{v}\} = \text{Null}(\lambda\boldsymbol{I}_n - \boldsymbol{A}).$$

Remarks

By definition, the geometric multiplicity of λ is equal to $\dim(E_\lambda)$.

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Let $\boldsymbol{A}$ be a square matrix of order n. Which of the following statements is/are true?

a. The characteristic polynomials of $\boldsymbol{A}$ and $\boldsymbol{A}^T$ are equal.

b. If λ is an eigenvalue of $\boldsymbol{A}$, then the algebraic multiplicity of λ as an eigenvalue of $\boldsymbol{A}$ is equal to the algebraic multiplicity of λ as an eigenvalue of $\boldsymbol{A}^T$.

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Question 2: Solution

a. $\det(x\boldsymbol{I}_n - \boldsymbol{A}) = \det((x\boldsymbol{I}_n - \boldsymbol{A})^T) = \det(x\boldsymbol{I}_n - \boldsymbol{A}^T)$.

b. If λ is an eigenvalue of $\boldsymbol{A}$, then the multiplicity of λ as a root of $\det(x\boldsymbol{I}_n - \boldsymbol{A})$ is equal to the multiplicity of λ as a root of $\det(x\boldsymbol{I}_n - \boldsymbol{A}^T)$, since $\det(x\boldsymbol{I}_n - \boldsymbol{A}) = \det(x\boldsymbol{I}_n - \boldsymbol{A}^T)$.

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which is the geometric multiplicity of λ as an eigenvalue of $\boldsymbol{A}^T$.

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which is the geometric multiplicity of λ as an eigenvalue of $\boldsymbol{A}^T$.

Question 3

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For each of the following matrices,

i. compute its characteristic polynomial and determine all its eigenvalues;

ii. for each eigenvalue λ, write down its algebraic and geometric multiplicities and find a basis for its associated eigenspace E_λ.

a. $\boldsymbol{A} = \begin{pmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{pmatrix}$.

b. $\boldsymbol{B} = \begin{pmatrix} 12 & 1 & 3 \\ -26 & 3 & -17 \\ -14 & -1 & -5 \end{pmatrix}$.

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Question 3a: Solution

$$\det(x\boldsymbol{I}-\boldsymbol{A}) = \det\begin{pmatrix} x-1 & 3 & -3 \\ -3 & x+5 & -3 \\ -6 & 6 & x-4 \end{pmatrix} = (x+2)^2(x-4).$$

Thus $\boldsymbol{A}$ has eigenvalues -2 and 4, with algebraic multiplicities 2 and 1 respectively.

$$-2\boldsymbol{I}-\boldsymbol{A} = \begin{pmatrix} -3 & 3 & -3 \\ -3 & 3 & -3 \\ -6 & 6 & -6 \end{pmatrix} \xrightarrow[\text{Jordan}]{\text{Gauss-}} \begin{pmatrix} 1 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow E_{-2} = \left\{ \begin{pmatrix} s-t \\ s \\ t \end{pmatrix} \middle| \, s,t \in \mathbb{R} \right\}.$$

Thus the geometric multiplicity of the eigenvalue -2 is 2, and $\left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}$ is a basis for E_{-2}.

$$4\boldsymbol{I}-\boldsymbol{A} = \begin{pmatrix} 3 & 3 & -3 \\ -3 & 9 & -3 \\ -6 & 6 & 0 \end{pmatrix} \xrightarrow[\text{Jordan}]{\text{Gauss-}} \begin{pmatrix} 1 & 0 & -\frac{1}{2} \\ 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow E_4 = \left\{ \begin{pmatrix} \frac{1}{2}t \\ \frac{1}{2}t \\ t \end{pmatrix} \middle| \, t \in \mathbb{R} \right\}.$$

Thus the geometric multiplicity of the eigenvalue 4 is 1, and $\left\{ \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ 1 \end{pmatrix} \right\}$ is a basis for E_4.

Question 3a: Solution

$$\det(x\boldsymbol{I}-\boldsymbol{A}) = \det\begin{pmatrix} x-1 & 3 & -3 \\ -3 & x+5 & -3 \\ -6 & 6 & x-4 \end{pmatrix} = (x+2)^2(x-4).$$

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Question 3a: Solution

$$\det(x\boldsymbol{I}-\boldsymbol{A})=\det\begin{pmatrix}x-1 & 3 & -3\\ -3 & x+5 & -3\\ -6 & 6 & x-4\end{pmatrix}=(x+2)^2(x-4).$$

Thus $\boldsymbol{A}$ has eigenvalues -2 and 4, with algebraic multiplicities 2 and 1 respectively.

$$-2\boldsymbol{I}-\boldsymbol{A}=\begin{pmatrix}-3 & 3 & -3\\ -3 & 3 & -3\\ -6 & 6 & -6\end{pmatrix}\xrightarrow[\text{Jordan}]{\text{Gauss-}}\begin{pmatrix}1 & -1 & 1\\ 0 & 0 & 0\\ 0 & 0 & 0\end{pmatrix}\Rightarrow E_{-2}=\left\{\begin{pmatrix}s-t\\ s\\ t\end{pmatrix}\;\middle|\;s,t\in\mathbb{R}\right\}.$$

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Question 3b: Solution

$$\det(x\boldsymbol{I}-\boldsymbol{B})=\det\begin{pmatrix} x-12 & -1 & -3 \\ 26 & x-3 & 17 \\ 14 & 1 & x+5 \end{pmatrix}=(x+2)(x-6)^2.$$

Thus $\boldsymbol{B}$ has eigenvalues -2 and 6, with algebraic multiplicities 1 and 2 respectively.

$$-2\boldsymbol{I}-\boldsymbol{B}=\begin{pmatrix} -14 & -1 & -3 \\ 26 & -5 & 17 \\ 14 & 1 & 3 \end{pmatrix}\xrightarrow[\text{Jordan}]{\text{Gauss-}}\begin{pmatrix} 1 & 0 & \frac{1}{3} \\ 0 & 1 & -\frac{5}{3} \\ 0 & 0 & 0 \end{pmatrix}\Rightarrow E_{-2}=\left\{\begin{pmatrix} -\frac{1}{3}t \\ \frac{5}{3}t \\ t \end{pmatrix}\,\middle|\, t\in\mathbb{R}\right\}.$$

Thus the geometric multiplicity of the eigenvalue -2 is 1, and $\left\{\begin{pmatrix} -\frac{1}{3} \\ \frac{5}{3} \\ 1 \end{pmatrix}\right\}$ is a basis for E_{-2}.

$$6\boldsymbol{I}-\boldsymbol{B}=\begin{pmatrix} -6 & -1 & -3 \\ 26 & 3 & 17 \\ 14 & 1 & 11 \end{pmatrix}\xrightarrow[\text{Jordan}]{\text{Gauss-}}\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{pmatrix}\Rightarrow E_{6}=\left\{\begin{pmatrix} -t \\ 3t \\ t \end{pmatrix}\,\middle|\, t\in\mathbb{R}\right\}.$$

Thus the geometric multiplicity of the eigenvalue 6 is 1, and $\left\{\begin{pmatrix} -1 \\ 3 \\ 1 \end{pmatrix}\right\}$ is a basis for E_6.

Question 3b: Solution

$$\det(x\boldsymbol{I}-\boldsymbol{B})=\det\begin{pmatrix}x-12 & -1 & -3\\ 26 & x-3 & 17\\ 14 & 1 & x+5\end{pmatrix}=(x+2)(x-6)^2.$$

Thus $\boldsymbol{B}$ has eigenvalues -2 and 6, with algebraic multiplicities 1 and 2 respectively.

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Question 4

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Let $\boldsymbol{A}$ be a square matrix of order n, and let λ be an eigenvalue of $\boldsymbol{A}$. Which of the following statements is/are always true?

a. For any $k \in \mathbb{Z}^+$, λ^k is an eigenvalue of $\boldsymbol{A}^k$.

b. If $\boldsymbol{A}$ is invertible, then $\frac{1}{\lambda}$ is an eigenvalue of $\boldsymbol{A}^{-1}$.



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Question 4: Solution

Solution.

Let $\boldsymbol{v}$ be an eigenvector of $\boldsymbol{A}$ associated with λ. Then $\boldsymbol{v} \neq \boldsymbol{0}$.

a. For any $k \in \mathbb{Z}^+$, λ^k is an eigenvalue of $\boldsymbol{A}^k$.
$\boldsymbol{A}^k\boldsymbol{v} = \boldsymbol{A}^{k-1}(\boldsymbol{A}\boldsymbol{v}) = \boldsymbol{A}^{k-1}(\lambda\boldsymbol{v}) = \lambda(\boldsymbol{A}^{k-1}\boldsymbol{v}) = \lambda(\boldsymbol{A}^{k-2}(\boldsymbol{A}\boldsymbol{v})) = \lambda^2\boldsymbol{A}^{k-2}\boldsymbol{v} = \cdots = \lambda^k\boldsymbol{v}$.
Thus $\boldsymbol{v}$ is an eigenvector of $\boldsymbol{A}^k$ associated with the eigenvalue λ^k.

b. If $\boldsymbol{A}$ is invertible, then $\frac{1}{\lambda}$ is an eigenvalue of $\boldsymbol{A}^{-1}$.
Since $\boldsymbol{A}$ is invertible, 0 is not an eigenvalue of $\boldsymbol{A}$, so $\lambda \neq 0$.
$\boldsymbol{v} = \boldsymbol{I}_n\boldsymbol{v} = \boldsymbol{A}^{-1}\boldsymbol{A}\boldsymbol{v} = \boldsymbol{A}^{-1}(\lambda\boldsymbol{v}) = \lambda\boldsymbol{A}^{-1}\boldsymbol{v}$.
Thus $\boldsymbol{A}^{-1}\boldsymbol{v} = \frac{1}{\lambda}\boldsymbol{v}$. This shows $\boldsymbol{v}$ is an eigenvector of $\boldsymbol{A}^{-1}$ associated with the eigenvalue $\frac{1}{\lambda}$.

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Next Lesson ...

We will explore **Sections 6.2 and 6.3**.
Please watch/review the video clips for these sections before then.